

CRADOCK, South Africa

WMO: 687440

Lat: 32.1675S

Lon: 25.6253E

Elev: 3038

StdP: 13.15

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.7	35.3	10.9	10.7	57.8	14.2	12.7	59.3	21.8	69.5	19.1	65.4	6.0	340	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.1	96.0	62.9	92.6	62.3	89.4	62.1	67.7	82.4	66.6	81.1	65.7	79.9	10.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	99.2	71.6	62.6	95.1	71.0	61.5	91.5	70.6	33.9	82.1	33.0	81.2	32.3	80.0	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
22.3	20.1	18.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			25.3	69.7	2.0	1.4	23.9	70.7	22.7	71.5	21.6	72.2	20.1	73.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	73.6	72.7	69.8	63.3	58.7	52.2	52.0	54.9	60.9	64.5	67.1	71.1
	DBStd	9.63	5.77	5.55	5.83	6.67	5.35	5.17	5.92	6.51	6.41	7.55	6.49	6.24
	HDD50	116	0	0	0	1	5	35	47	23	2	2	1	0
	HDD65	1781	6	7	21	109	206	384	403	319	157	102	50	17
	CDD50	4986	730	635	614	401	273	101	109	176	329	451	514	653
	CDD65	1179	271	223	170	59	10	0	0	7	35	86	113	205
	CDH74	11412	2512	1836	1411	528	125	7	12	118	461	1028	1279	2095
	CDH80	4750	1151	770	548	152	14	0	0	23	139	423	531	999
Wind	WSAvg	8.3	9.5	8.5	8.1	6.9	6.4	7.2	7.2	7.3	8.3	9.5	10.0	10.4
Precipitation	PrecAvg	11.9	1.4	1.7	1.8	1.2	0.4	0.4	0.5	0.5	0.3	1.1	1.5	1.6
	PrecMax	16.7	5.4	5.3	5.6	3.1	2.4	1.7	3.0	2.2	2.8	4.2	4.8	5.8
	PrecMin	5.8	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.0	1.3	1.3	1.2	0.8	0.6	0.4	0.7	0.6	0.6	1.1	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.8	98.7	94.6	88.8	81.8	74.2	76.0	83.7	89.3	95.3	96.3	100.3
		MCWB	64.3	63.6	62.4	60.5	56.4	51.6	50.5	53.5	56.3	60.1	60.8	61.5
	2%	DB	94.6	92.9	90.4	84.1	77.4	70.4	72.1	77.3	84.1	89.7	91.2	94.9
		MCWB	64.3	64.2	63.1	59.7	54.5	50.0	49.7	51.8	55.3	58.6	59.5	62.4
	5%	DB	90.5	88.3	86.2	80.0	73.7	67.5	68.6	72.4	79.3	84.5	86.5	90.3
		MCWB	63.9	64.3	62.7	58.4	53.9	48.8	48.7	50.6	54.5	57.5	59.0	61.9
	10%	DB	86.3	84.3	82.1	75.6	69.8	63.9	64.3	67.8	74.5	79.5	81.8	85.4
		MCWB	63.7	64.3	61.9	57.5	52.9	47.4	47.3	48.9	53.0	56.9	58.6	61.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	69.9	67.7	64.0	59.0	53.7	52.7	57.0	60.2	63.4	65.1	67.1
		MCDB	85.4	85.2	84.5	77.0	71.9	68.7	70.6	71.5	79.5	82.2	82.5	84.6
	2%	WB	67.6	67.9	66.3	62.4	57.4	51.7	51.0	54.8	58.1	61.7	62.9	65.6
		MCDB	81.8	82.2	82.2	75.3	71.1	66.3	67.5	69.3	75.3	79.8	81.4	82.8
	5%	WB	66.6	66.7	64.9	61.0	55.9	50.1	49.7	53.0	56.5	60.3	61.5	64.5
		MCDB	80.9	80.0	79.0	74.6	68.4	63.4	65.4	65.9	72.0	77.7	79.3	81.4
	10%	WB	65.6	65.8	63.7	59.3	54.5	48.5	48.1	51.1	54.8	58.6	60.1	63.3
		MCDB	79.9	78.3	76.5	70.8	66.1	61.1	61.1	64.0	70.3	75.3	76.9	79.1
Mean Daily Temperature Range	5% DB	MDBR	25.1	23.7	24.1	23.4	23.2	23.0	24.2	24.4	25.8	26.5	26.6	26.2
		MCDBR	31.7	30.3	29.7	28.9	28.0	27.9	29.0	31.2	32.9	34.5	34.9	35.0
		MCWBR	7.7	7.7	8.6	9.4	11.0	13.2	13.3	13.4	12.4	10.9	10.1	8.8
	5% WB	MCDBR	25.3	24.1	25.3	24.2	23.1	24.8	26.0	25.2	27.8	29.3	28.5	26.6
MCWBR		7.7	7.8	8.5	8.4	9.8	12.3	12.7	10.7	11.1	10.5	10.1	8.6	
Clear-Sky Solar Irradiance	taub	0.337	0.342	0.328	0.301	0.283	0.270	0.270	0.312	0.338	0.348	0.327	0.331	
	taud	2.518	2.506	2.519	2.575	2.600	2.605	2.594	2.452	2.367	2.382	2.493	2.525	
	Ebn at Noon	318	310	303	296	286	283	288	287	295	304	319	321	
	Edh at Noon	36	35	32	27	23	22	23	30	37	39	36	36	
All-Sky Solar Radiation	RadAvg	2347	2131	1804	1397	1106	978	1063	1327	1719	2055	2328	2490	
	RadStd	119	116	102	61	43	54	60	80	104	104	121	147	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page